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JAIPUR NATIONAL UNIVERSITY

End Semester/Year Examination December -2025

Name of School:	School of Engineering & Technology
Name of Programme / Course:	B.Tech (Computer Science Engineering / AIML / CY / DATA SCIENCE)
Semester/Year:	V / III
Paper Code:	BTECSCO502T22
Title of Paper:	Operating Systems
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

1. This question paper consists of two sections, Section-A and Section-B.
2. Section-A consists of seven questions and
3. Section-B consists of six questions.

Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question

Question Number	SECTION - A	Marks Allotted	Bloom Level	Course Outcome
Q.1	Define an Operating System. Explain the various services provided by an OS.	4	1	1
Q.2	Discuss the differences between Monolithic, Layered, and Microkernel architectures of operating systems with examples.	4	2	2
Q.3	Illustrate the different states of a process using a neat process state transition diagram.	4	3	2
Q.4	Differentiate between process and thread. Explain the advantages of	4	4	3

multithreading in modern OS design.

~~Q.5~~ Describe Peterson's Solution for mutual exclusion and discuss how it avoids race conditions. 4 3 4

~~Q.6~~ Evaluate the advantages and limitations of different page replacement algorithms used in virtual memory management. 4 5 5

~~Q.7~~ Write short notes on Virtual Machine and Microkernel Architecture. 4 1 1

SECTION - B

Q.1 Explain in detail the structure of an operating system. Compare UNIX and Windows OS architectures, highlighting key design differences, system calls, and process management approaches. 10 2 1

Q.2 Define and discuss CPU Scheduling. Explain the working of FCFS, SJF, and Round Robin scheduling algorithms with suitable examples. Evaluate their performance based on turnaround and waiting time. 10 5 2

Q.3 Analyze the critical section problem and describe how semaphores and monitors solve synchronization issues. Support your explanation with examples from the producer-consumer or reader-writer problems. 10 4 3

Q.4

Q.5

Q.6

4	Q.4	Discuss the concept of deadlocks in operating systems. Explain Banker's Algorithm in detail and illustrate with a suitable example how deadlock avoidance is ensured.	10	3	4
5	Q.5	Describe the memory management techniques in an OS. Discuss paging and segmentation in detail, and compare their advantages and limitations in terms of efficiency and protection.	10	5	5
1	Q.6	Explain the I/O subsystem structure of an operating system. Discuss the role of device drivers, interrupt handlers, and disk scheduling algorithms in achieving efficient input/output operations.	10	1	3
1					
2					